

The BFGS quasi Newton method

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$$\text{GD: } x_{t+1} = x_t - \eta \cdot \textcircled{I} \cdot \nabla f(x_t) \quad \xleftarrow[\text{minimize?}]{} \quad \text{Newton: } x_{t+1} = x_t - \eta \cdot \textcircled{\nabla^2 f(x_t)^{-1}} \cdot \nabla f(x_t)$$

Key goal: approximate the Hessian without requiring expensive computation

Key idea: build the approximation recursively, using curvature information along the generated trajectory.

Reference: Nocedal & Wright - Numerical Optimization

Any quadratic approximation defines a direction

$$f(x_k + d) \approx m_k(d) = f(x_k) + \nabla f(x_k)^T d + \frac{1}{2} d^T B_k d$$

$$d_k = -B_k^{-1} \nabla f(x_k)$$

$$\text{QN: } x_{k+1} = x_k - \eta \underset{\substack{\uparrow \\ \text{chosen by B.T.L.S.}}}{B_k^{-1}} \cdot \nabla f(x_k)$$

→ Update B_k iteratively

Desiderata :

Ⓐ Would like: $B_k^{-1} \nabla f(x_k)$ to be "easy" to compute.

Ⓑ $m_k(d)$ to capture curvature information of f along the trajectory ✓

Key Idea: match curvature along trajectory

BFGS: $f(x_k + d) \approx m_k(d)$, $f(x_{k+1} + d) \approx m_{k+1}(d)$

We require: m_{k+1} matches the gradient of f at the last two iterations.

$$(i) \quad \left. \nabla m_{k+1} \right|_{d=0} = \nabla f(x_{k+1})$$

$$(ii) \quad \left. \nabla m_{k+1} \right|_{d=-\eta_k d_k} = \nabla f(x_k)$$

$$\begin{aligned} & \left[\begin{aligned} x_{k+1} &= x_k + \eta_k d_k \\ \Leftrightarrow x_k &= x_{k+1} - \eta_k d_k \end{aligned} \right. \end{aligned}$$

The Secant Equation

$$m_{k+1}(d) = f(x_{k+1}) + \nabla f(x_{k+1})^T d + \frac{1}{2} d^T B_{k+1} d$$

$$\nabla m_{k+1}(d) = \nabla f(x_{k+1}) + B_{k+1} d$$

(i) $\nabla m_{k+1}(0) = \nabla f(x_{k+1})$ — automatically satisfied — ✓

(ii) $\nabla m_{k+1}(-\eta_k d_k) = \underbrace{\nabla f(x_{k+1}) - \eta_k B_{k+1} d_k}_{\text{want}} = \underbrace{\nabla f(x_k)}_{\text{want}}$

$$\eta_k B_{k+1} d_k = \nabla f(x_{k+1}) - \nabla f(x_k)$$

Recall: $x_{k+1} = x_k + \eta_k d_k \iff \eta_k d_k = x_{k+1} - x_k$

$$B_{k+1} \underbrace{(x_{k+1} - x_k)}_{s_k} = \underbrace{\nabla f(x_{k+1}) - \nabla f(x_k)}_{y_k}$$

Secant Equation: $B_{k+1} s_k = y_k$

Defining the Hessian approximation via optimization

$$\left[\begin{array}{l} B_{k+1} = \operatorname{argmin}: \|B - B_k\| \\ B = B^T \\ B s_k = y_k \end{array} \right. \quad \begin{array}{l} \textcircled{1} \quad B_{k+1} \text{ close to } B_k \\ \textcircled{2} \quad B_{k+1} \text{ is symmetric} \\ \textcircled{3} \quad B_{k+1} \text{ sat. secant equ.} \end{array}$$

Each choice of norm gives a different B_{k+1}
satisfying secant eq and thus a different QN method

$$\text{BFGS:} \quad H_k = B_k^{-1}$$

$$\textcircled{1} \quad H_{k+1} \text{ is close to } H_k$$

$$\textcircled{2} \quad H_{k+1} \text{ is symmetric}$$

$$\textcircled{3} \quad H_{k+1} \text{ satisfies secant equ; } s_k = H_{k+1} y_k$$

Instead: directly approximating the Inverse Hessian

$$\begin{aligned} H_{k+1} &= \arg \min: \|H - H_k\| \\ \text{st: } & H = H^T \\ & H y_k = s_k \end{aligned}$$

$$\text{Recall: } \|A\|_F^2 = \sum_{i,j} (A_{ij})^2 \quad (\text{Frobenius norm})$$

$$\|A\|_W^2 = \|W^{1/2} A W^{1/2}\|_F^2 \quad (\text{Weighted Frobenius Norm})$$

$$\text{BFGS uses: } W = \int_0^1 \nabla^2 f(x_k + t y_k) dt$$

$$\underline{\text{Then}}: H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T$$

$$\rho_k = \frac{1}{y_k^T s_k}$$

Check: H_{k+1} satisfies the secant equation.

The BFGS update direction and algorithm

BFGS Algorithm: for $k=1, \dots, T$ (or $\|\nabla f(x_k)\| < \varepsilon$)

$$d_k = -H_k \nabla f(x_k)$$

$$x_{k+1} = x_k + \eta_k d_k = x_k - \eta_k H_k \nabla f(x_k)$$

(η_k from BTLB)

$$s_k = x_{k+1} - x_k$$

$$y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$$

$$H_{k+1} = \text{— from prev slide —}$$

Back Tracking Line Search for BFGS

sufficient decrease:

$$f(x_k + \eta_k d_k) \leq f(x_k) + \alpha_1 \cdot \eta_k \nabla f(x_k)^T d_k$$

we will also require a second condition

$$\nabla f(x_k + \eta_k d_k)^T d_k \geq \alpha_2 \cdot \nabla f(x_k)^T d_k$$

$$0 < \alpha_1 < \alpha_2 < 1.$$

Limited Memory BFGS

Computation required for BFGS: $O(n^2)$

Nevertheless: H_k $n \times n$ dense matrix

Observation: H_{k+1} is a rank 2 update from H_k
only need: s_k, y_k

L-BFGS